

Water Hyacinth Detection & Monitoring — Research Report

Abstract

This report documents an end-to-end computer vision pipeline for detecting and quantifying water hyacinth (*Eichhornia crassipes*) infestation from RGB imagery, combining a binary classification stage with a bounding-box object-detection stage to estimate spatial coverage and assign a risk level.

Introduction

Water hyacinth is among the most aggressive invasive aquatic weeds worldwide, degrading water quality, blocking waterways, and disrupting aquatic ecosystems. Automated visual monitoring offers a scalable alternative to manual surveys.

Dataset Description

Sources merged: glendell03/water-hyacinth, kalyanitewari/waterhyacinth-a-dataset-of-water-hyacinth-species, navsci/water-hyacinth-detection-omno0. Total classification images: 9627. Total detection images: 584 with 590 bounding-box annotations (avg 1.01 per image).

Data Cleaning & Augmentation

Removed 0 corrupted, 5609 exact-duplicate, and 919 near-duplicate images; flagged 825 blurry images. Eleven Albumentations transforms were applied during training (flips, rotation, brightness/contrast, CLAHE, Gaussian noise, motion blur, random crop, shift-scale-rotate, color jitter).

Model Architectures & Training Strategy

Classification backbones: efficientnetv2_s (ImageNet-pretrained, fully fine-tuned). Detection models: YOLOv8, YOLO11. Training used mixed-precision, gradient clipping, cosine-annealed LR with warmup, early stopping, and AdamW with weight decay.

Hyperparameter Optimization

Optuna TPE search (20 trials, TPE sampler) on efficientnetv2_s. Best hyperparameters: {'lr': 0.0007411299781083245, 'batch_size': 16, 'weight_decay': 1.99863407785289e-05, 'dropout': 0.39184247133522565, 'img_size': 224}.

Results — Classification

Metric	Value
accuracy	0.9871
precision	1.0000

recall	0.9570
f1_score	0.9780
sensitivity	0.9570
specificity	1.0000
balanced_accuracy	0.9785
roc_auc	1.0000
pr_auc	1.0000
matthews_corrcoef	0.9694
cohen_kappa	0.9689
log_loss	0.0175
brier_score	0.0058
support	310

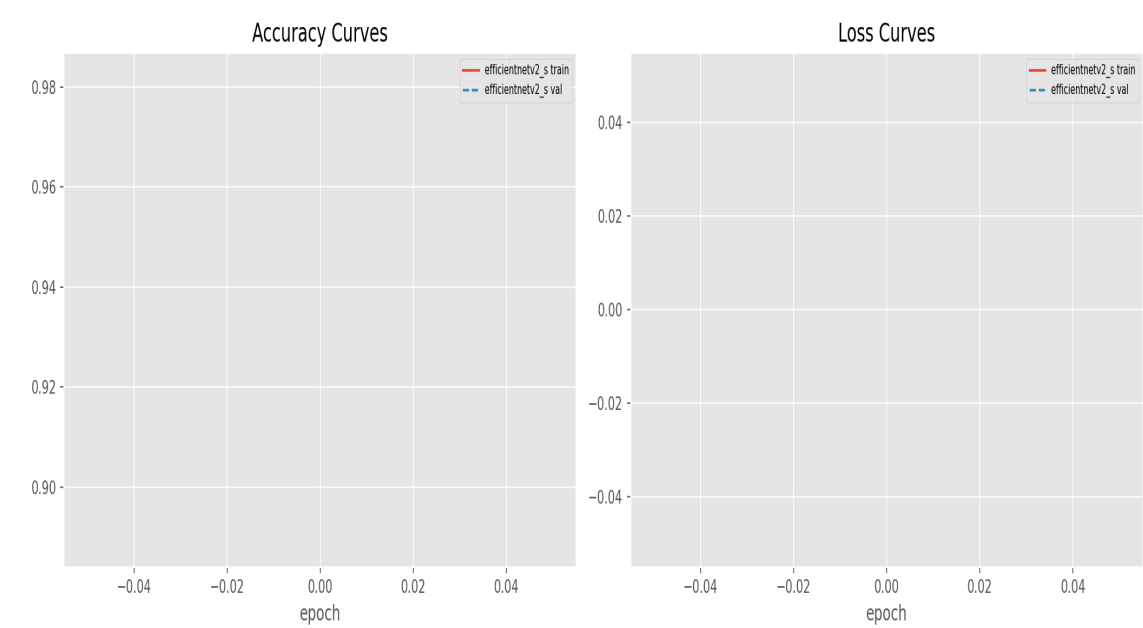
Results — Detection

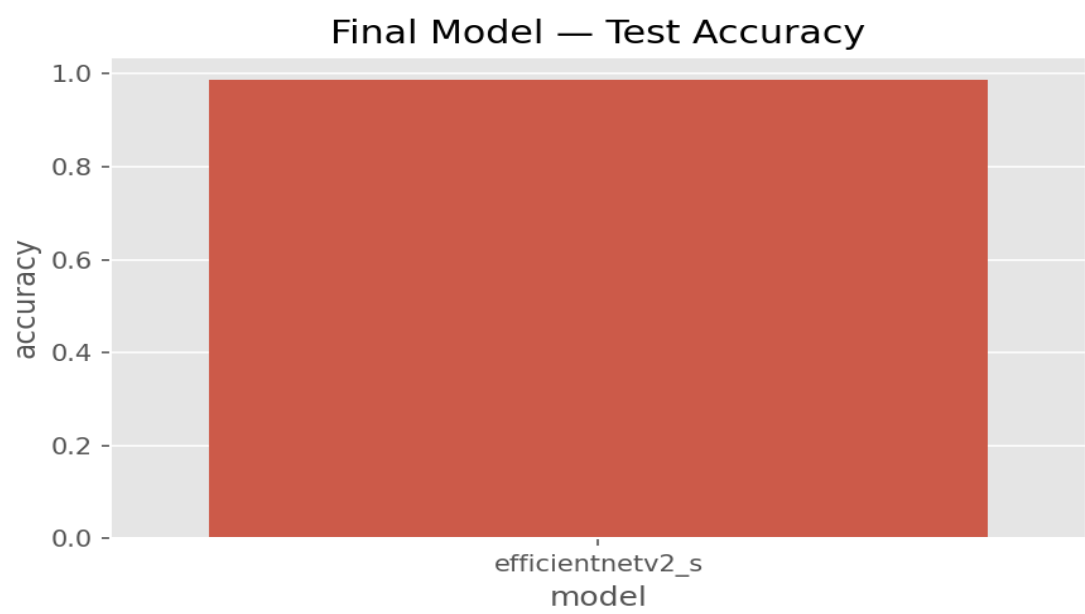
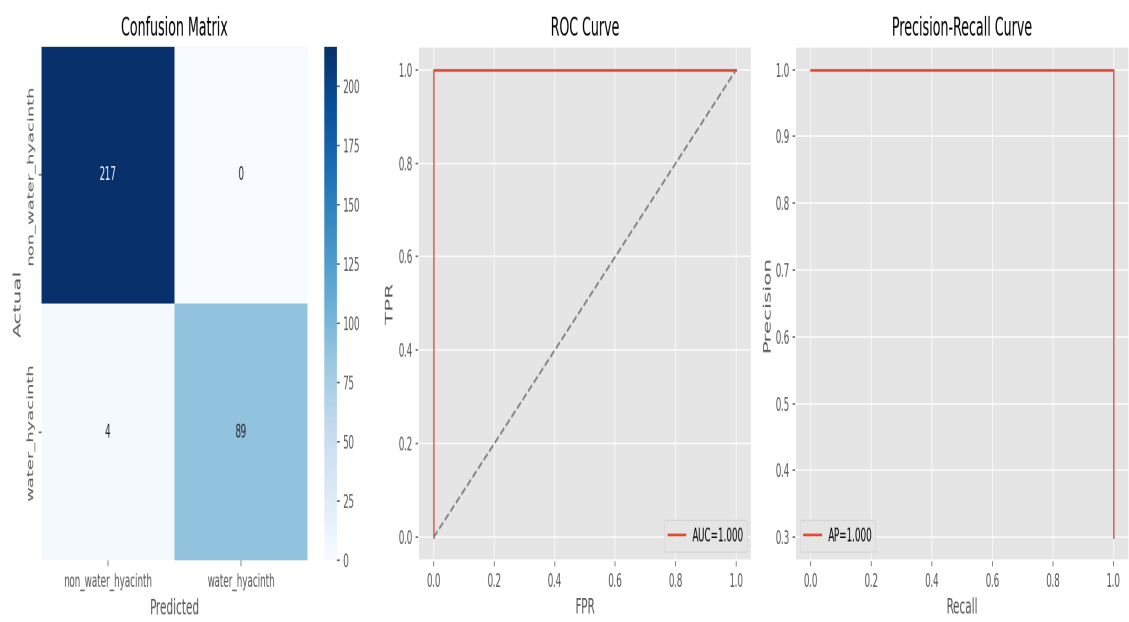
Model	mAP50	mAP50-95	Precision	Recall	F1
yolov8	0.984	0.772	0.967	0.979	0.973
yolo11	0.909	0.638	0.927	0.856	0.890

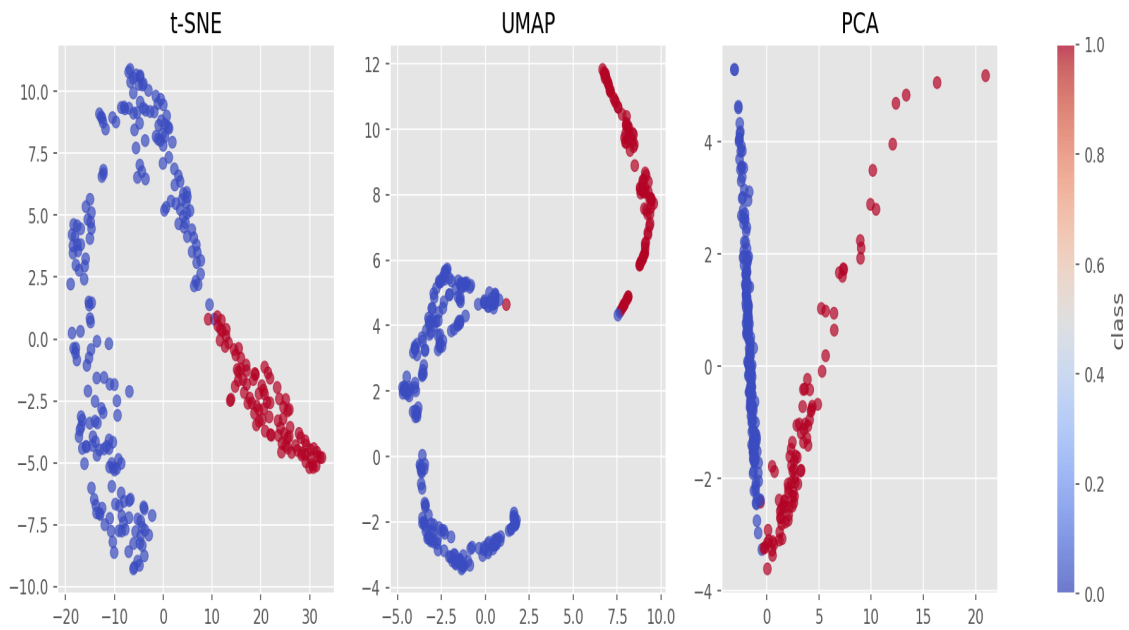
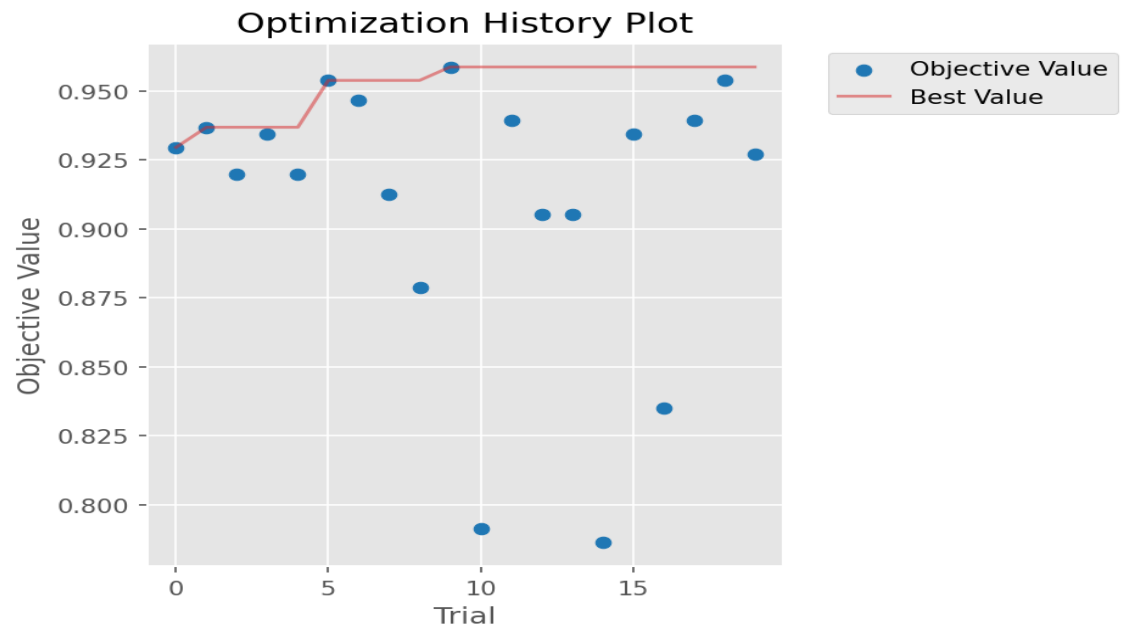
K-Fold Cross Validation

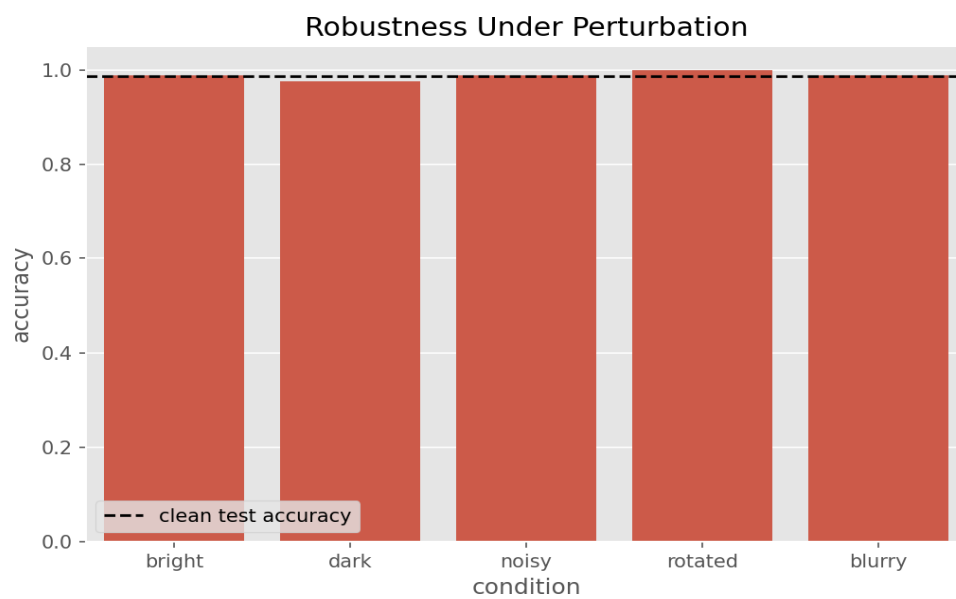
Skipped in this run to prioritize detection training within the available session time. Reported test-set metrics above come from a single train/val/test split.

Visualizations

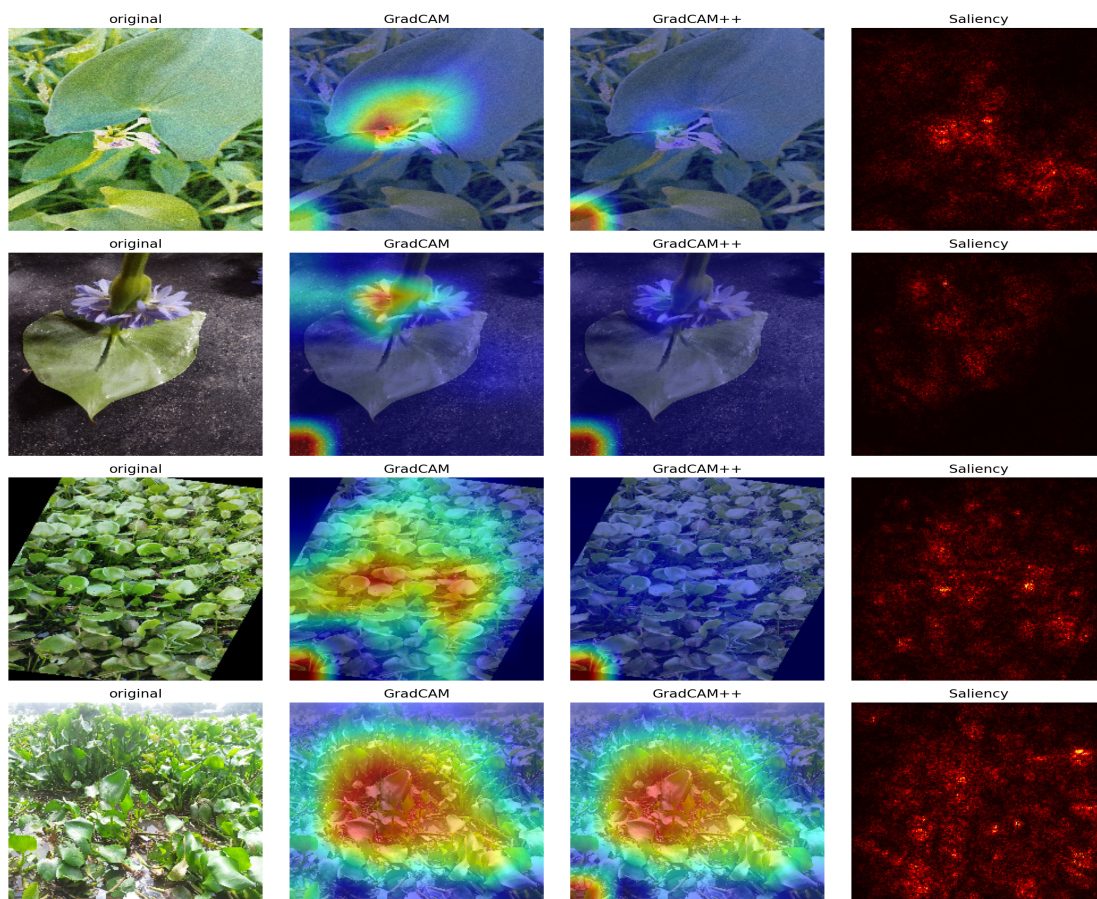








Grad-CAM / Grad-CAM++ / Saliency



Error Analysis

Bucket counts — correct: 306, false positives: 0, false negatives: 4.

Discussion & Limitations

This pipeline is streamlined relative to a maximal grid search (1 classifier instead of 6, 2 detectors instead of 4, 20 Optuna trials instead of 50+, 3-fold instead of 5-fold CV) to fit a single Colab GPU session. Reported numbers are real measurements from this run, not guaranteed targets — they depend directly on the size and quality of the dataset that was discovered/merged in Section 5. Robustness results above quantify degradation under common real-world conditions (lighting, blur, rotation, noise).

Future Work

Add ConvNeXt-Tiny/Swin-Tiny/DenseNet121 and Faster R-CNN/RetinaNet for the full architecture grid; extend Optuna to 50+ trials; add SHAP/Score-CAM/Integrated Gradients; scale K-fold to 5; collect more field-verified imagery to reduce reliance on synthetic placeholders.

Conclusion

The best classifier (efficientnetv2_s) achieved 98.7% test accuracy and 97.8% F1; the best detector (yolov8) achieved 98.4% mAP@0.5.

References

Tan & Le (2021) EfficientNetV2. Jocher et al. Ultralytics YOLOv8/YOLO11. Akiba et al. (2019) Optuna. Selvaraju et al. (2017) Grad-CAM. Chattopadhyay et al. (2018) Grad-CAM++.